

Mahfuz Raihan

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SUMMARY

Highly motivated and results-driven ML Engineer with a strong background in developing and deploying cutting-edge AI solutions. Expertise includes building product recommender systems and advanced RAG-based AI Chatbots featuring Agentic AI capabilities. Skilled in NLP with experience in LangChain, scikit-learn, LangGraph, and Pytorch. Proven ability to establish robust data pipelines, manage ML experimentation, and implement model versioning with MLflow. Experienced in model deployment using Azure ML with CI/CD practices. Eager to contribute to a collaborative team that drives significant business impact.

TECHNICAL SKILLS

Languages	: Python
ML/DL Frameworks	: PyTorch, Scikit-learn, OpenCV, NLTK, SpaCy, torchtext
Generative AI/LLM	: LangChain, Ollama
MLOps & Cloud	: Azure ML, MLflow, Docker, GitHub Actions, Git, PyTest
Data & Analytics	: Pandas, Dask, NumPy, SQLAlchemy
Vector DB	: FAISS, ChromaDB, pgvector
Database and BD Tools	: PostgreSQL, MySQL, MongoDB, SQL Server, GCP BigQuery
Visualization	: Matplotlib, Plotly, Google Looker Studio
API Frameworks	: FastAPI
AI Agentic Workflow	: LangGraph, CrewAI, OpenClaw, Multi-Agent Systems, Human-in-the-Loop (HITL)
Web Frameworks	: Flask, Streamlit, Django

PROFESSIONAL EXPERIENCE

Machine Learning Engineer

May 2025 – Present

Brain Station 23 – MetLife

Dhaka, Bangladesh

- Developed production-grade personalized recommendations engine, increases lead engagement 17% and reduces training time 70%
- Built scalable and robust data pipelines, reduces time 80%
- Implemented MLOps best practices using **MLflow** for experiment tracking and model versioning, reducing deployment friction.
- Architected an Enterprise RAG-based AI Chatbot using **LangChain** and **LangGraph** for internal knowledge management.
- Automated document data extraction using **Azure Document Intelligence**, improving text processing efficiency.
- Deployed ML services on **Azure ML** with integrated **CI/CD** and performance monitoring to detect model drift.

Jr. Data Scientist

Nov 2023 – April 2025

Tallykhata / TallyPay (Progoti Systems Limited)

Dhaka, Bangladesh

- Developed a predictive churn model to identify high-value "Premium" customers, enabling targeted retention strategies.
- Designed 10+ dynamic real-time KPI dashboards using **BigQuery** to monitor transactional and user growth metrics.
- Optimized SQL functions in **PostgreSQL** to analyze complex customer behavioral journeys for business intelligence.
- Monitored model performance using **MLflow** and **TensorBoard**, performing hyperparameter tuning to increase F1-scores from 0.77 to 0.85
- Automated stakeholder reporting by building a Python-based delivery system connected to the central data warehouse.

Trainee Jr. Data Scientist

June 2023 – Oct 2023

Tallykhata / TallyPay (Progoti Systems Limited)

Dhaka, Bangladesh

- Worked on **Tallykhata Data warehouse**, developed **task automation** using Python.
- Implemented **advanced analytics queries** to support decision-making for various product stakeholders.
- Created and optimized **SQL queries** to analyze complex customer behavioral patterns and identify key business insights.
- Collaborated with cross-functional teams to understand business requirements and deliver data-driven solutions.
- Built **ML models** to identify and prevent fraudulent transactions. F1 score: 0.70 to 0.86

PROJECTS/OPEN SOURCE

NetOps-AI Agent: Autonomous SOC | OpenClaw, Llama 3.3, PyTorch, FastAPI, SpaCy

GitHub

- Developed an autonomous AI agent for real-time cyber threat detection and filtering using zero-shot learning.
- Integrated a Human-in-the-Loop (HITL) dashboard for administrative oversight of AI-driven security responses.
- Utilized **OpenClaw's** flexible LLM framework to orchestrate threat analysis, enabling high-performance zero-shot detection.
- Leveraged **FastAPI** to build a scalable backend API and implemented **PyTorch** for efficient model inference.
- Designed a production-ready system architecture, ensuring reliability and scalability for enterprise security operations.

Intelligent Data Assistant – Agentic RAG System <i>LangChain, LangGraph, FAISS, Azure OpenAI</i>	GitHub
• Built a state-machine driven RAG pipeline with self-reflection capabilities to verify and retry retrieval failures. • Implemented vectorized document processing for diverse file types (PDF, XLSX) using FAISS embeddings. • Integrated Azure OpenAI for advanced text understanding and information extraction capabilities. • Generate dynamic Plotly charts from user-uploaded datasets. • Designed a production-ready system architecture, ensuring reliability and scalability for enterprise security operations.	
Small Scale AI Search Engine: Text to Image Search <i>PyTorch, OpenAI CLIP, FastAPI, ChromaDB</i>	GitHub
• Retrieval based text to image search engine using OpenAI CLIP Model. • Implemented vector search using ChromaDB for efficient image retrieval. • Integrated FastAPI to build a scalable backend API and implemented PyTorch for efficient model inference. • Find the most relevant images by calculating the cosine similarity between text and image embeddings. • Explored the use of OpenAI CLIP Model to understand the semantic content of both text and images.	
Bengali Vehicle Plate Detection (YOLO) <i>PyTorch, YOLOv10, OpenCV</i>	
• Achieved 97% accuracy in detecting and extracting Bengali text from vehicle number plates in real-time environments. • Use for Production in any system where this is needed.	

EDUCATION

Islamic University, Bangladesh	2016 – 2022
<i>B.Sc. in Information and Communications Technology</i>	<i>Kushtia, Bangladesh</i>
Kushtia Government College	2012 – 2014
<i>Higher Secondary Certificate(HSC) – Science</i>	<i>Kushtia, Bangladesh</i>
Kushtia Zilla School	2007 – 2012
<i>Secondary School Certificate(SSC) – Science</i>	<i>Kushtia, Bangladesh</i>

CERTIFICATIONS AND LEARNING

Machine Learning for Production	Coursera
Deep Learning with PyTorch: Generative AI	Coursera
IBM Data Science Professional	Coursera
• Data Science Methodology – Coursera • Data Visualization using Python –Coursera	
Python for Everybody Specialization	
• Programming for Everybody – Coursera • Python Data Structures – Coursera	
Kaggle certificate	
• Data Analysis with pandas – Kaggle	

AWARD AND HONOR

1st Runner Up, Project Showcase (SUST ICT Fest 2019)	April 2019
Champion (Group Team), Central Science Fair 2019, Islamic University, Bangladesh	February 2019
Champion (Group Team), Central Science Fair 2018, Islamic University, Bangladesh	March 2018